



Using formulas

Teacher resource

Question 1

The formula to find the circumference of a circle is $C = 2\pi r$, where r is the radius.

Which expression represents the total distance around three identical circles?

- ☐ $2\pi r \div 3$
- ☐ $3\pi r$
- ☐ $6\pi r$
- ☐ $2\pi r + 3$

Answer: C

Skill: Students manipulate simple formulas

Additional questions

1. Write an expression that represents the distance around five identical circles.
2. Write an expression that represents the distance around a semi-circle.

Question 2

The formula to convert a temperature in degrees Celsius ($^{\circ}\text{C}$) to a temperature in degrees Fahrenheit ($^{\circ}\text{F}$) is given by

$$\frac{9}{5}C + 32 = F$$

If the Fahrenheit temperature is 100°F , what is the temperature in degrees Celsius?

Answer: 37.8°C

Skill: Students substitute values and solve an equation.

Additional questions

1. What is the temperature in degrees Fahrenheit if the Celsius temperature is 32°C ?
2. What is the temperature in degrees Celsius if the Fahrenheit temperature is 0°F ?

Question 3

A finance company uses the following series of formulae to calculate the growth of an investment over a number of years:

Year 1 = initial investment

Year 2 = Year 1 $\times$ 1.15

Year 3 = Year 2 $\times$ 1.15

Year 4 = Year 3 $\times$ 1.15

By how much would an initial investment of \$2000 increase over a period of four years?

Answer: \$1041.75

Skill: Students substitute into a formula

Additional questions

1. How many years will it take for the initial investment to double?
2. An initial investment grew to \$1983.75 after three years. What was the initial investment?

Question 4

The volume, V , of a sphere with a radius of r , is given by the formula

$$V = \frac{4}{3} \pi r^3$$

Which expression gives the radius of a sphere in terms of the volume?

O $\sqrt[3]{\frac{4V}{3\pi}}$

O $\frac{3}{4} \sqrt{\frac{V}{\pi}}$

O $\sqrt[3]{\frac{3V}{4\pi}}$

O $\frac{4}{3} \sqrt{\frac{V}{\pi}}$

Answer: C

Skill: Students rearrange an algebraic equation involving index notation.

Additional questions

1. Thirty spheres, each with a volume of 40 cm^3 , were melted to form one large sphere. How many times larger is the radius of this sphere than the radius of the original small spheres?
2. How many spheres, each with a volume of 40 cm^3 , would be needed to make one large sphere that has a radius four times the radius of the smaller spheres?

Student activity: Using formulas

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